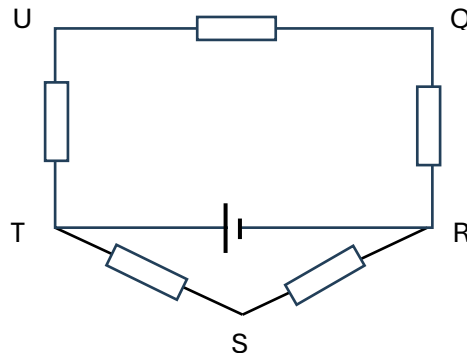


- 22 Five resistors, each of $50\ \Omega$, are connected in a loop as shown below. A $10.0\ \text{V}$ battery of zero internal resistance is connected across TR. What is the p.d. between U and R, V_{UR} ?



- A** $10.0\ \text{V}$ **B** $8.7\ \text{V}$ **C** $6.7\ \text{V}$ **D** $3.7\ \text{V}$

Ans: C

TSR is irrelevant to the calculation.

$$V_{QR} = \frac{R_{QR}}{R_{TU} + R_{QR}} \times 10 = \frac{50 + 50}{50 + 50 + 50} \times 10 = 6.67\ \text{V}$$

- 23 Two long straight wire X and Y are placed perpendicular to each other at a distance d apart. A